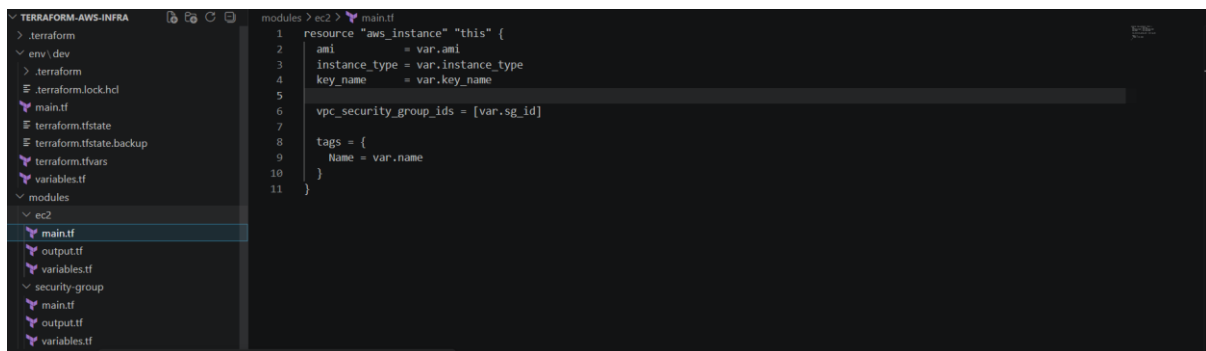
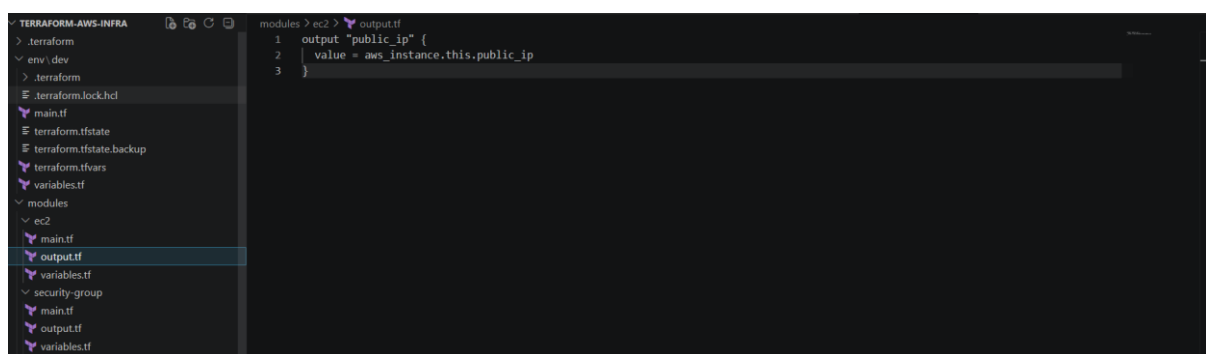


CREATE A MODULE IN TERRAFORM

1. In terraform you can create new folder and then create folder EC2, Security Group, Env and Dev.
2. In every folder you have to create a file with name main.tf, variables.tf, output.tf.
3. In Env and Dev you have create a file with name main.tf, variables.tf and terraform.tfvars.
4. After that don forget to terraform plan and then terraform apply.
5. Then go to AWS dashboard and go to EC2 you will see EC2 you created earlier.
6. And try the going to th website with click public IP IPV4.
7. You will see the website not working right now, you have to install httpd first.
8. We will SSH the server , in here iam using Git Bash.
9. Just Type : sudo yum install httpd > sudo systemctl start httpd > sudo enable httpd.
10. Refresh the website you we will see a small type “its work”



(EC2 MAIN.TF)



(EC2 OUTPUT.TF)

```
TERRAFORM-AWS-INFRA
> .terraform
> env\dev
> terraform
  terraform.lock.hcl
  main.tf
  terraform.tfstate
  terraform.tfstate.backup
  terraform.tfvars
  variables.tf
modules
  ec2
    main.tf
    output.tf
    variables.tf
  security-group
    main.tf
    output.tf
    variables.tf

modules > ec2 > variables.tf
1 variable "ami" {
2   | type = string
3   }
4
5 variable "instance_type" {
6   | type = string
7   }
8
9 variable "key_name" {
10  | type = string
11  }
12
13 variable "sg_id" {
14   | type = string
15   }
16
17 variable "name" {
18   | type = string
19   }
```

(EC2 VARIABLES.TF)

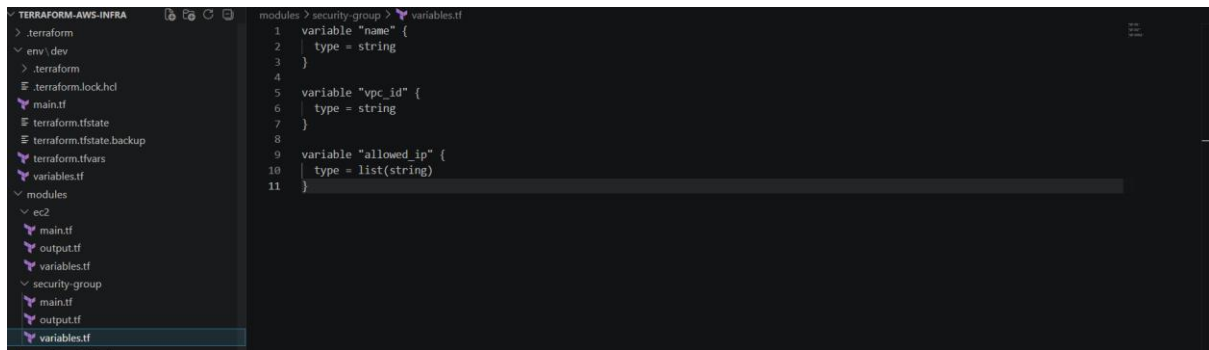
```
modules > security-group > main.tf
1 resource "aws_security_group" "this" {
2   name     = var.name
3   vpc_id   = var.vpc_id
4
5   ingress {
6     description = "SSH"
7     from_port   = 22
8     to_port     = 22
9     protocol    = "tcp"
10    cidr_blocks = var.allowed_ip
11  }
12
13   ingress {
14     description = "HTTP"
15     from_port   = 80
16     to_port     = 80
17     protocol    = "tcp"
18     cidr_blocks = var.allowed_ip
19   }
20
21   egress {
22     from_port   = 0
23     to_port     = 0
24     protocol    = "-1"
25     cidr_blocks = ["0.0.0.0/0"]
26   }
27
28   tags = {
29     Name = var.name
30   }
31 }
```

(Security Group MAIN.TF)

```
TERRAFORM-AWS-INFRA
> .terraform
> env\dev
> terraform
  terraform.lock.hcl
  main.tf
  terraform.tfstate
  terraform.tfstate.backup
  terraform.tfvars
  variables.tf
modules
  ec2
    main.tf
    output.tf
    variables.tf
  security-group
    main.tf
    output.tf
    variables.tf

modules > security-group > output.tf
1 output "sg_id" {
2   | value = aws_security_group.this.id
3   }
```

(Security Group OUTPUT.TF)



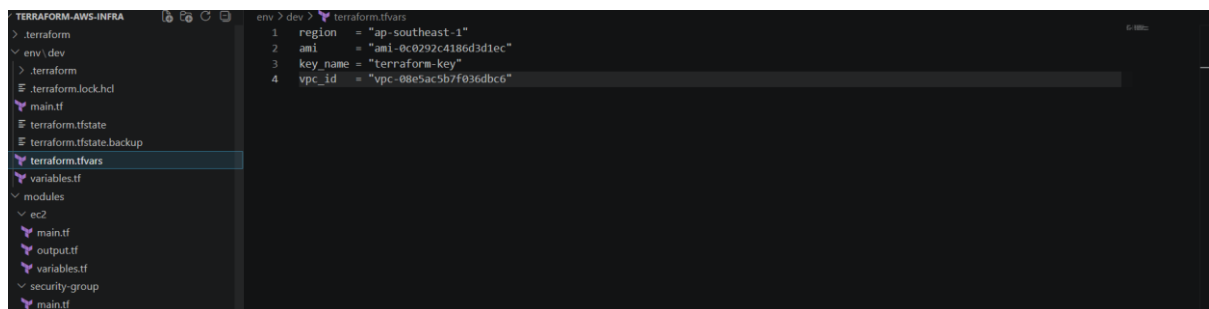
```
modules > security-group > variables.tf
1 variable "name" {
2   | type = string
3   |
4   |
5   variable "vpc_id" {
6     | type = string
7     |
8     |
9   variable "allowed_ip" {
10    | type = list(string)
11    |
12   |
```

(Security Group VARIABLES.TF)



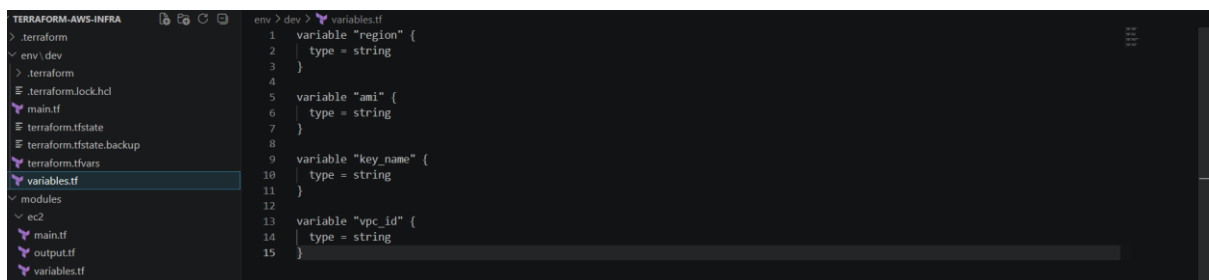
```
1 provider "aws" {
2   | region = var.region
3   |
4   |
5   module "sg" {
6     | source = "../../modules/security-group"
7     | name = "dev-sg"
8     | vpc_id = var.vpc_id
9     | allowed_ip = ["0.0.0.0/0"]
10    |
11    |
12   module "ec2" {
13     | source = "../../modules/ec2"
14     | ami = var.ami
15     | instance_type = "t3.micro"
16     | key_name = var.key_name
17     | sg_id = module.sg.sg_id
18     | name = "dev-server"
19     |
```

(ENV/DEV MAIN.TF)



```
env > dev > terraform.tfvars
1 region = "ap-southeast-1"
2 ami = "ami-0c9292c4186d3d1ec"
3 key_name = "terraform-key"
4 vpc_id = "vpc-08e5ac5b7f036dbc6"
```

(ENV/DEV TERRAFORM.TFVARS)



```
env > dev > variables.tf
1 variable "region" {
2   | type = string
3   |
4   |
5   variable "ami" {
6     | type = string
7     |
8     |
9   variable "key_name" {
10    | type = string
11    |
12    |
13   variable "vpc_id" {
14     | type = string
15     |
```

(ENV/DEV VARIABLES.TF)

EC2 > Instances

EC2

- Dashboard
- AWS Global View
- Events
- Instances
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - Capacity Manager
- Images
 - AMIs
 - AMI Catalog
- Elastic Block Store

Instances (1/2) info

Connect Instance state Actions Launch instances

Find Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IP
dev-server	i-04ef64d9745b858ff	Running	t3.micro	3/3 checks passed	View alarms	ap-southeast-1c	ec2-13-214-216-12
web-server	i-08ce7cb655399a6b9	Stopped	t3.micro	-	View alarms	ap-southeast-1b	-

i-04ef64d9745b858ff (dev-server)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary info

Instance ID: i-04ef64d9745b858ff

Public IPv4 address: 13.214.216.12 | open address

Private IPv4 addresses: 172.31.1.75

Instance state: Running

Public DNS: ec2-13-214-216-12.ap-southeast-

13.214.216.12

This page isn't working right now

13.214.216.12 didn't send any data.

Refresh

Microsoft Edge

It works!

```
ec2-user@ip-172-31-1-75:~$  
  
Installed:  
apr-1.7.5-1.amzn2023.0.4.x86_64  
apr-util-1.6.3-1.amzn2023.0.2.x86_64  
apr-util-ldap-1.6.3-1.amzn2023.0.2.x86_64  
apr-util-openssl-1.6.3-1.amzn2023.0.2.x86_64  
generic-logos-httpd-18.0.0-12.amzn2023.0.3.noarch  
httpd-2.4.66-1.amzn2023.0.1.x86_64  
httpd-core-2.4.66-1.amzn2023.0.1.x86_64  
httpd-filesystem-2.4.66-1.amzn2023.0.1.noarch  
httpd-tools-2.4.66-1.amzn2023.0.1.x86_64  
libbrotli-1.0.9-4.amzn2023.0.2.x86_64  
mailcap-2.1.49-3.amzn2023.0.3.noarch  
mod_http2-2.0.27-1.amzn2023.0.3.x86_64  
mod_lua-2.4.66-1.amzn2023.0.1.x86_64  
  
Complete!  
[ec2-user@ip-172-31-1-75 ~]$ sudo systemctl start httpd  
[ec2-user@ip-172-31-1-75 ~]$ sudo systemctl enable httpd  
Created symlink /etc/systemd/system/multi-user.target.wants/httpd.service → /usr/lib/systemd/system/httpd.service.  
[ec2-user@ip-172-31-1-75 ~]$
```